Lab 5 - Logistic Regression and K-Nearest Neighbors

Binary Classification Using a Real Dataset

Aim

To introduce binary classification using an actual dataset. Part A develops Logistic Regression from the data to a complete working classifier. Part B asks students to apply K-Nearest Neighbors (KNN) to the same problem after the KNN method is explained during the laboratory session.

Learning Outcomes

- Identify the features and binary target of a classification dataset.
- Explain how Logistic Regression converts a linear score into a probability using the sigmoid function.
- Interpret predicted probabilities and convert them to class labels using a decision threshold.
- Prepare data correctly for classification using stratified train-test splitting and feature standardization.
- Train and evaluate a Logistic Regression classifier using a confusion matrix, accuracy, precision, recall and F1-score.
- Apply KNN to the same dataset, investigate the effect of the number of neighbours k, and compare its classification performance with Logistic Regression.

Dataset

Both parts use the Breast Cancer Wisconsin (Diagnostic) dataset available through scikit-learn. It contains 569 observations and 30 numerical input features calculated from digitized images of breast-mass cell nuclei.

- Examples of input features include mean radius, mean texture, mean perimeter, mean area and related measurements.
- Original scikit-learn target: 0 = malignant, 1 = benign.
- For this lab we remap the target to 1 = malignant and 0 = benign, so the positive class represents malignant cases.

Part A - Logistic Regression

Part A is completed in the notebook and is used during the laboratory explanation. All Logistic Regression concepts are introduced through the actual Breast Cancer dataset.

A1. Load and understand the classification problem

1. Load the dataset and inspect its shape, feature names and class distribution.
2. Create the input matrix X using all 30 features.
3. Remap the target so that malignant = 1 and benign = 0.
4. Visualize two real features against each other to see that the two classes overlap and cannot be separated perfectly by a simple rule.

A2. From a linear score to a probability

Logistic Regression first forms a linear score:

$$z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ldots + \theta_n x_n$$

The score z is converted to a value between 0 and 1 using the sigmoid function:

$$p = 1 / (1 + e^{-z})$$

In this lab, p is interpreted as the model's estimated probability that the observation belongs to class 1 (malignant). The notebook shows this using an actual test observation by comparing the model's decision score, the manually calculated sigmoid value and predict_proba().

A3. Probability to class decision

Using the default threshold of 0.5:

$$p \geq 0.5 \rightarrow \text{class } 1 \quad | \quad p < 0.5 \rightarrow \text{class } 0$$

The notebook displays actual test cases together with their malignant probability and final predicted class, so the probability-to-decision process can be observed directly.

A4. Prepare and train the model

5. Split the data into training and testing sets using stratify=y so that class proportions are maintained.
6. Fit StandardScaler only on the training features and use the same scaler to transform the test features.
7. Fit sklearn LogisticRegression on the standardized training data.
8. Generate probabilities and class predictions for the test set.

A5. Evaluate the working classifier

Evaluate the completed model using:

- Confusion matrix
- Accuracy
- Precision
- Recall
- F1-score

Because malignant is defined as the positive class, recall answers an important question: "Of the actual malignant cases, how many did the classifier identify as malignant?"

In-class Demonstration - Understanding KNN Before Part B

Before Part B, KNN is introduced using a small two-feature dataset. The example makes the distance calculation, neighbour selection and voting process visible before KNN is applied to the 30-feature Breast Cancer dataset.

- Plot a small labelled two-class dataset and one new observation.
- Calculate Euclidean distances from the new observation to the training observations.
- Sort the observations by distance and identify the nearest neighbours.
- Compare majority voting using k = 1, 3 and 5.
- Observe that changing k can change the prediction.
- Compare KNN with Logistic Regression before starting the student exercise.

Logistic Regression vs KNN

Logistic Regression learns a global linear classification model and produces probabilities. KNN is an instance-based method that makes local decisions from nearby training observations.

Aspect	Logistic Regression	KNN
Decision idea	Global equation + sigmoid	Neighbour distance + voting
Boundary	Linear by default	Can be nonlinear
Best suited to	Probability modelling, interpretable global relationship	Small/medium local-similarity problems
Large datasets	Prediction is efficient	Prediction can become expensive
High dimensions	Often suitable with regularization	Can suffer from curse of dimensionality

Part B - Student Exercise: Apply KNN to the Same Dataset

KNN will be explained separately during the laboratory session. In Part B, complete the TODO cells without copying the prepared Logistic Regression pipeline directly. Start again from the raw Breast Cancer dataset.

9. Load the dataset and recreate X and the remapped target y (malignant = 1, benign = 0).
10. Create a stratified training/testing split using random_state = 42.
11. Fit a StandardScaler using only the training features and transform both training and test features.
12. Train KNeighborsClassifier with k = 5.
13. Predict the test-set classes.
14. Create the confusion matrix and calculate accuracy, precision, recall and F1-score.
15. Repeat KNN for several odd k values and observe how the classification performance changes with k.
16. Compare Logistic Regression and KNN on the same test data. At minimum, compare their test accuracy; you may also use precision, recall, F1-score, confusion matrices or other outputs if useful.
17. Present the final comparison in any clear form you prefer, such as a table, chart, printed summary, short notebook explanation or a combination. There is no required presentation format.

Why scaling is important for KNN

KNN decides which observations are neighbours using distance. If one feature has a much larger numerical scale than another, it can dominate the distance calculation. Standardization places the numerical features on comparable scales before distances are calculated.

Submission

Submit one file only:

- Completed Lab 5 notebook (.ipynb) with all Part B TODO cells completed, the final model comparison included, and outputs visible.

Suggested filename: REGNO_Lab05.ipynb